

Dataset description

16-QAM Optical Constellation Samples Under Man-In-The-Middle Attacks

Type Synthetic Data

Version 1.0

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URL <https://doi.org/10.34810/data2661>

1.1 DATA GENERATION DESCRIPTION

This section describes the following data:

- **Reference Optical Constellation Samples (no attack):** This dataset contains optical constellation samples for lightpaths of distance ranging from 80km to 2000km, which allows emulating the operation of a large number of lightpath configurations. This dataset has been extracted and pre-processed from the data publicly available in the “Optical Constellation Analysis (OCATA)” dataset (<https://doi.org/10.34810/data146>)
- **Altered Samples with MitM attack Combining Tampering and Evasion:** This dataset contains optical constellation samples originally belonging to the previous dataset after the application of the HOCUS methodology and algorithms to perform a sophisticated MitM attack combining tampering + evasion.

1.1.1 Reference Optical Constellation Samples (no attack)

This section summarizes the dataset generation procedure of the reference optical constellation samples, focusing on those aspects that are more relevant for the understanding of the attacks in further datasets. For additional details not included here (as well as alternative reference optical constellation measurements), please access dataset and descriptions in <https://doi.org/10.34810/data146>.

A MATLAB-based simulator of a coherent WDM system has been developed to generate IQ constellations for a 16QAM@64GBd signal under different physical path characteristics. Assuming 100 GHz channel spacing and full spectrum occupancy, signal samples containing 2,048 symbols and shaped by a root-raised cosine filter with a 0.06 roll-off-factor are generated at the Tx side. Then, the signal is propagated through standard single mode fiber spans, characterized by optimal power of -1 dBm, attenuation factor of 0.21 dB/km, dispersion parameter of 16.8 ps/nm/km, and nonlinear parameter of 1.14 1/W/km. Spans are modeled by solving the nonlinear Schrödinger equation using the well-known split-step Fourier method, whereas ideal inline optical amplification is modelled as erbium-doped fiber amplifiers with a noise figure of 4.5 dB, introducing linear noise. Finally, at the Rx, a DSP block able to perform ideal chromatic dispersion compensation and phase recovery is considered.

A sequence of up to 25 spans without intermediate ROADMs between Tx and Rx is configured. Spans have a fixed length of 80km; thus, constellation samples with a total distance ranging from 80km to 2000 km have been generated.

1.1.2 Altered Samples with MitM attack Combining Tampering and Evasion

This section summarizes the procedure to perform the attack that combines tampering and evasion, named *Hacking Optical Constellations through re-Shaping (HOCUS)*, whose detailed algorithms and performance are detailed in the publication “**Optical Layer and Quantum Digital Twins for Enhancing Secure Autonomous Network Operation**”, in Proceedings of International Conference in Transparent Optical Networks (ICTON), 2025 ([10.1109/ICTON67126.2025.11125305](https://doi.org/10.1109/ICTON67126.2025.11125305)).

HOCUS performs an evasion attack by modifying I-Q symbols fingerprints in a way that optical constellation samples generated by an adversary (Mallory) and reaching the victim (Bob) resemble the expected for the original signal. Thus, HOCUS receives source optical symbols (hereafter denoted as x_a) with total accumulated length L_a and produces modified target optical symbols (x_b) with a dispersion around the optical constellation point that resembles length L_b . With this general notation, we cover different situations, e.g., the adversary gets control of

destination site and perform both tampering and evasion in the same site, or take physical control on an intermediate site and logical control of destination site. Note that, in the cases mentioned before, $L_a < L_b$. However, there are cases where $L_a > L_b$, e.g., if Mallory could only gain physical access to a site that is not in the original route of the lightpath. In this case, the attacker could silently reroute the lightpath so it crosses the desired intermediate site. As a consequence of this action, L_b might be shorter than L_a .

HOCUS is defined as a parametric function that requires from a set of coefficients ϑ to be trained before performing any re-shaping. Assuming 16QAM signals, $\vartheta = \{\vartheta_1, \dots, \vartheta_{16}\}$, where ϑ_i initially contains the expected bi-variate Gaussian distribution of constellation point i for both L_a and L_b lengths. Without loss of generality, we assume that Mallory can access monitoring data repositories, obtain true optical constellation samples from all or part of the lightpaths in operation, and train mean (μ) and covariance matrix (Σ) of bi-variate Gaussian distributions before performing MitM attacks. However, samples for distances L_a and/or L_b might not be available, so polynomial interpolation is used to estimate missing coefficients from the available lightpath lengths.

Once μ and σ for lengths L_a and L_b are obtained, transformation matrices W are computed and added to ϑ . Σ is semi-definite positive and hence, a decomposition $\Sigma = P \cdot D \cdot P^*$ exists, being P a unitriangular matrix, D a diagonal one and P^* the conjugate transpose of P . Then, transformation matrix $W_i(L_a, L_b)$ for constellation point i is computed:

$$W_i(L_a, L_b) = P_i(L_b) \cdot D_i^{\frac{1}{2}}(L_b) \cdot P_i^*(L_b) \cdot P_i(L_a) \cdot D_i^{-\frac{1}{2}}(L_a) \cdot P_i^*(L_a)$$

Individual symbol transformation can then be easily performed. Let us denote $x_a(t)$ as the source symbol processed by HOCUS at time t . Assuming that it belongs to constellation point i , the target symbol $x_b(t)$ is obtained by applying the transformation:

$$x_b(t) = W_i(L_a, L_b) \cdot (x_a(t) - \mu_i(L_a)) + \mu_i(L_b)$$

Although the transformation can be applied to all constellation points, in this dataset we only focus on 4 of them that are widely used in optical constellation analysis, which are those belonging to the expected centroids: $[-3+3j]$, $[1+1j]$, $[-1-1j]$, and $[1-3j]$.

1.2 DATA FILE CONVENTIONS

The dataset contains two folders, one for the reference data described in Section 1.1.1 and another for the attack data in Section 1.1.2.

1.2.1 Reference folder

It contains 25 files, named with the convention "source_n", where n represents the lightpath distance in km from transceiver to receiver. Each file contains 50 samples of 2048 symbols each, stored in a tabular format. The columns of the table are:

- **Sample_Id**: Numerical identifier of the sample, relative to the file (ranging from 1 to 50)
- **Symbol_Id**: numerical identifier of the symbol, relative to the sample (ranging from 1 to 2048)
- **Real**: The real component (I axis or x-axis) of the symbol
- **Imag**: The imaginary component (Q axis or y-axis) of the symbol

1.2.2 Altered folder

It contains 25 files, named with the convention “source_n”, where n represents the distance from the tampering site to the receiver. Since we assume that evasion performed by HOCUS is done at the receiver site, this distance in the filename equals L_a . Each file contains a variable number of samples of 2048 symbols each, stored in a tabular format. The columns of the table are:

- **Source_Distance:** Distance in km from tampering site to receiver (a.k.a, L_a)
- **Target_Distance:** Distance in km from transceiver to receiver (a.k.a, L_b)
- **Sample Id:** Numerical identifier of the sample, relative to the file
- **Symbol Id:** numerical identifier of the symbol, relative to the sample
- **Real:** The real component (I axis or x-axis) of the symbol
- **Imag:** The imaginary component (Q axis or y-axis) of the symbol

1.3 SUPPORTING CODE

To facilitate the understanding of data organization and processing, an illustrative code is provided. The code is a plain Python file that receives L_a and L_b parameters. Note that it will work if and only if:

- L_a is in the range [80, 1920] km, in steps of 80 km.
- L_b is in the range [160, 200] km, in steps of 80 km, and $L_b > L_a$

The code properly reads the data from the files, corresponding to distances L_a and L_b and plots all the available samples (symbols). Figure 1 illustrates the case where $L_a = 160$ km and $L_b = 800$ km. The figure at the left shows the reference signal that receives no attacks. The figure at the right shows the effect of the attacks. On the one hand, the selected constellation points $[-3+3i]$, $[1+1i]$, $[-1-1i]$, and $[1-3i]$ perform combined tampering and evasion, whereas the remaining constellation points only perform tampering. In this way, it can be seen how combined attack effectively resembles well the shape of original reference signals. On the contrary, tampering without evasion makes symbols' dispersion much more different than reference one (less dispersion), so it is easy to detect in case of having prior knowledge on expected dispersion.

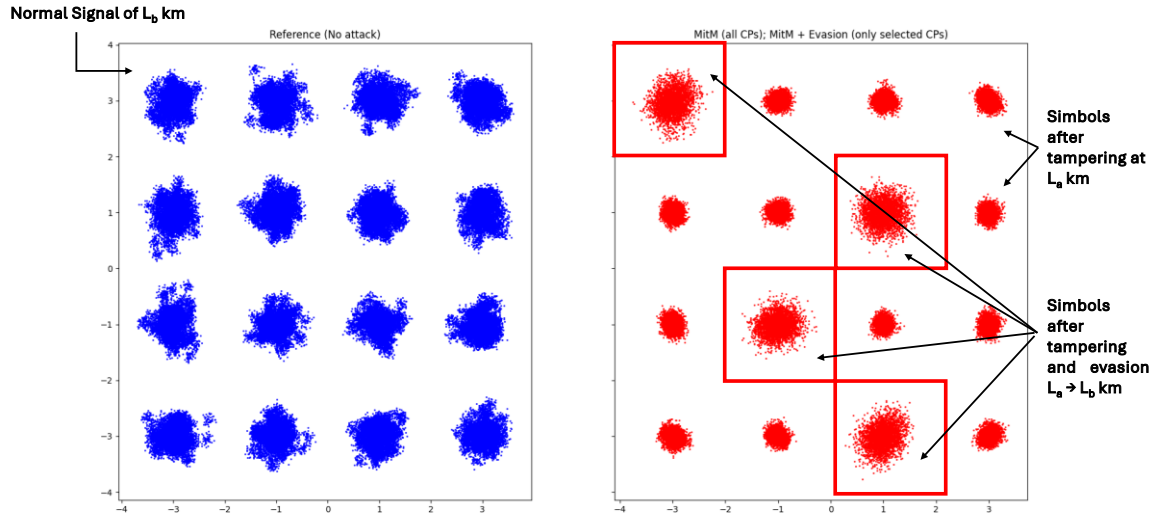


Figure 1: Output of illustrative data processing code (with additional notes)

1.4 ACKNOWLEDGEMENTS

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